

# **From Wealth to Well-Being: GDP, Human Development and the Evolving Meaning of Economic Progress; An Empirical Study.**

*Abhishikta Ganguly*  
*BSc Semester 3*

## **ABSTRACT**

Economic development is perhaps one of the most important standards against which the progress of nations is assessed in the modern world. To this end, the Gross Domestic Product (GDP) has emerged as the dominant indicator of economic performance, shaping how governments, institutions and societies evaluate national progress. Yet, an increase in aggregate output might not necessarily imply a corresponding improvement in the quality of people's lives. This paradox leads us to a very fundamental question: Is GDP growth sufficient to capture a country's holistic development in the long run?

The paper explores this distinction through the HDI approach which shifts the focus of development from the resources people have, to the freedoms and opportunities available to them to pursue the lives they value. In an attempt to broaden the already existing framework, the Human Development Index (HDI) represents an important alternative. By incorporating crucial parameters like health, education and standard of living alongside economic indicators, HDI provides a multidimensional picture of national progress. However, it too remains an incomplete measure as it cannot fully capture important factors like inequality, environmental sustainability, security, social relationships or subjective welfare.

The paper therefore does not seek to replace GDP with HDI or another single measure. Instead, it asks whether the objective of development should be understood as producing more, or as expanding people's opportunities to live better lives. The paper will complement this theoretical discussion with a comparative empirical analysis of GDP and human-development indicators in the country, examining whether economic growth translates into broader human development.

## **1. Introduction**

In a world that is evolving faster than ever, it has become imperative for economic development to be more holistic. Previously the GDP per capita was used as the sole yardstick to measure a country's economic progress. But over the years, it has not only proved to be insufficient, but also very restrictive in its approach. Constructed against the backdrop of the Great Depression, the 'GDP' was seen as a means to monitor the US economy's decline and subsequent reconstruction.

Simon Kuznets, the creator of the GDP, is said to have famously remarked: "The welfare of a nation [can] scarcely be inferred from a measurement of national income." Among various kinds of criticism of the GDP per capita, is that it only measures the market value of goods and services for the market and government and excludes many non-market goods and services such as housework or unpaid care. The second criticism is more concerning as the GDP fails to incorporate Inequalities within its framework. Third, and perhaps most fundamental, economic growth, depending on the way it is achieved, may be at the expense of social welfare, leisure time, equity and environmental quality (Stiglitz et al., 2009, pp. 12–3, 21, 85–91). In spite of all the critiques against this approach, it continued to be the predominant framework to discuss theoretical assumptions and empirical works, for a very long time. Keeping in mind, the caveats issued by the giants, it was important to look for newer approaches to measuring economic welfare.

## **2. HDI- A brief historical over-view**

While the concept of “capability” as a means to achieving holistic economic development is an age-old one, it is only recently that the theory has been formulated as a quantifiable approach. The HDI, as we know it today, was first introduced in 1990. It was initially designed as an unweighted arithmetic average of three dimensions: income, education, and life expectancy (UNDP, 1990). The HDI was created to emphasize that people and their capabilities should be the ultimate criteria for assessing the development of a country, instead of economic growth being a standalone parameter for development. (UNDP, HDR). Its importance became prominent as more and more experts came to the consensus that a holistic picture of a country’s level of human development requires analysis of other indicators and information metrics.

But this alone was not sufficient, as the growth of such a metric raised concerns over all the “dimensions” that could be included within the framework of the HDI. This led to several variations of the same concept over the next few decades, and is still a matter of deliberation for contemporary economists. In the next section we shall take a look at the three different constructions of the HDI (labelled HDI1, HDI2, HDI3) before beginning with the time series analysis of the same.

## 2.1 FIRST RECONSTRUCTION OF HDI:

In our first reconstruction of the HDI (later labelled HDI1) we have used a framework that very closely resembles the one employed by Crafts(1997). Crafts (1997b) used the original UNDP methodology to estimate the HDI for 16 countries in 1870, an additional 8 countries in 1913, and subsequently for 15 more countries in 1950, 1970, and 1992. In his earliest HDI methodology, he calculated using the following formula:

$$HDI_1 = \frac{1}{3} I_{SCH} + \frac{1}{3} I_{LE} + \frac{1}{3} L_Y$$

The following HDI reconstruction includes 3 broad “dimensions” of welfare. LE stands for Life Expectancy (the health metric), SCH stands for Mean Years of Schooling (the education metric), and Y denotes Income level (measured in GDP per Capita terms). Income was discounted above the poverty line using a piecewise concave function (Anand & Sen, 2000). Each variable ( $I_X$ ) is normalized between 0 and 1 as follows:

$$I_X = (x - x_{min}) / (x_{max} - x_{min}),$$

where  $x_{max}$  and  $x_{min}$  stand for the maximum and minimum value also referred to as “goalposts” (Rijpma et al, 2025), set to a 0–1 range of variable x.

It must be noted that in all the subsequent studies conducted by Crafts, the HDI reconstruction used Literacy Rate as the Education metric. However, due to abrupt discontinuities in the literacy rate data of India in this time period, we have chosen “Mean Years of Schooling” as the Education metric. While this can be perceived as a diversion from the original version used by Crafts, it is reassuring to know that the following modification has been adopted in various reports previously. The piecewise concave function captures the diminishing marginal utility of wealth by progressively compressing income segments that exceed a baseline threshold, reflecting that higher income yields decreasing returns to human development.

## 2.2 SECOND RECONSTRUCTION OF HDI

The modern-day studies on HDI are mostly rooted in the framework proposed by the UNDP. It varies from the original version proposed by Crafts, in that, it uses a Geometric mean of the Dimensions, as opposed to Arithmetic mean. From 2014 onwards, major studies have used this method. All these studies have varied from each other in their choice of “Dimensions”, or the transformations applied on these variables. (Rijpma et al, 2025) In most of these contributions, a geometric mean was used and literacy was substituted with Mean years of schooling (SCH) (Rijpma et al, 2025) Following a similar pattern, we calculate HDI2 using the following formula:

$$HDI_2 = I_{SCH}^{\left(\frac{1}{3}\right)} I_{LE}^{\left(\frac{1}{3}\right)} L_Y^{\left(\frac{1}{3}\right)}$$

Similar to the previous reconstruction, LE represents Life Expectancy (the health metric), SCH stands for Mean Years of Schooling (the education metric), and Y denotes Income level (measured in GDP per Capita terms).

## 2.3 THIRD RECONSTRUCTION OF HDI

Following UNDP’s HDI methodology of 2010 as the starting point, Leandro Prados de la Escosura used a geometric average to aggregate the indicators. He further applied a logarithmic transformation on the per capita income, while keeping the dimensions same. The key methodological innovation used was a convex transformation for life expectancy and education, which can be expressed for a social indicator  $x$  and its goalposts  $x_{max}$  and  $x_{min}$  as:

$$\frac{\log(x_{max} - x_{min}) - \log(x - x_{min})}{\log(x_{max} - x_{min})}$$

Prados de la Escosura (2022) argues that this approach should be applied based on contemporary quality-adjusted education and life expectancy data, which show that quality in these indicators improves more at higher levels of attainment.<sup>12</sup> This results in the following formula, with  $F_x$  as the convex transformation of education and life expectancy:

$$HDI_3 = F_{SCH}^{\frac{1}{3}} F_{LE}^{\frac{1}{3}} L_Y^{\frac{1}{3}}$$

## 3. STATISTICAL METHODOLOGY

The objective of the empirical analysis is to examine whether economic growth is associated with subsequent improvements in human development beyond the income dimension itself. The central research question is therefore: Is growth in GDP per capita associated with subsequent improvements in non-income human development in India, and is this relationship robust to alternative constructions of non-income HDI?

The analysis uses annual Indian data covering 33 observations. The economic variable is GDP per capita measured in constant 2015 US dollars. Human development is represented

through three alternative reconstructions of the HDI and their corresponding non-income counterparts. Health is represented by life expectancy, while education is represented by mean years of schooling and expected years of schooling.

Three alternative HDI constructions are employed (as described in the previous section). For each construction, the income component is removed to produce three non-income measures. (We obtain an index which we call NIHDI or Non-Income Human Development Index). Removing the income component does not generate a unique human-development index, and different aggregation functions can produce different numerical outcomes. The three measures therefore allow the analysis to examine whether the empirical relationship is sensitive to the construction of the dependent variable. It recognises that there is no unique way of constructing an index once the income dimension is deliberately excluded from HDI. Second, it permits a robustness assessment: if the estimated relationship between economic growth and non-income human development remains qualitatively similar across the three constructions, the result is less likely to be the result of a particular index formulation. The modern UNDP-style measure,  $NIHDI_2$ , is treated as the principal specification, while  $NIHDI_1$  and  $NIHDI_3$  are used as alternative constructions, throughout the procedure. Instead of using the GDP per capita term, as it is, we apply a log transformation.

$$X = \ln(GDPpc_t)$$

The logarithmic transformation is both economically and statistically significant. Economically, proportional changes in income are more meaningful across substantially different income levels than equal absolute changes in dollars, and the relationship between income. More importantly, the improvements in human-development outcomes are not linear in raw GDP per capita (Statistical calculations have already testified for that). The logarithm also compresses the scale of the GDP variable. For the final specification, the first difference of log GDP per capita,

$$\Delta \ln(GDPpc_t) = \ln(GDPpc_t) - \ln(GDPpc_{t-1})$$

The empirical strategy proceeds in stages. At the very first step, the levels relationship between GDP per capita and non-income human development is examined. Second, the time-series properties of the variables and the residual behaviour of the baseline model are assessed. Third, a time-trend specification is estimated to determine whether the GDP association remains distinguishable from the common evolution of both variables over time. Finally, the analysis moves from levels to annual changes and introduces a one-year lag in GDP growth to allow for the possibility that improvements in health and education respond to economic expansion with a delay.

The final specification is:

$$\Delta NIHDI_t = \alpha + \beta \Delta \ln(GDPpc_{t-1}) + \varepsilon_t$$

This specification examines whether GDP-per-capita growth in the previous year is associated with subsequent improvement in non-income human development. The use of a lag does not establish a causal mechanism; rather, it captures temporal precedence and allows the possibility that economic expansion takes time to be reflected in health- and education-related outcomes.

## **4. Statistical Analysis and Calculations**

### **4.1 Baseline levels regression**

The initial model estimates the association between the level of non-income human development (NIHDI) and the logarithm of GDP per capita. The equational form of the model thereby obtained is:

$$NIHDI2_t = \beta_0 + \beta_1 \ln(GDPpc_t) + u_t.$$

Ordinary least squares estimates the coefficient vector by minimizing the residual sum of squares,

$$RSS = \sum_{t=1}^n (NIHDI2_t - \widehat{NIHDI2}_t)^2.$$

For the principal  $NIHDI_2$  specification, the estimated equation is:

$$\widehat{NIHDI2}_t = -0.7643 + 0.1762 \ln(GDPpc_t). \\ (Author's own calculations)$$

The coefficient on log GDP per capita is positive and statistically significant, and the model produces a huge  $R^2$  value. This can apparently indicate a very large degree of association between the dependent and explanatory variables.

$$R^2 = 0.987.$$

Thus, approximately 98.7% of the variation in the observed level of  $NIHDI_2$  is accounted for by the fitted levels relationship within the sample. The correlation between log GDP per capita and  $NIHDI_2$  is approximately:

$$r = 0.993.$$

The corresponding correlations for the alternative constructions are also extremely high. This testifies for the results obtained through our analysis.

$$r(\ln G, DPpcNIHDI_1) \approx 0.993, \\ r(\ln G, DPpcNIHDI_3) \approx 0.994.$$

These results establish an extremely strong raw association, but they are not interpreted as establishing that GDP independently determines human development. Both GDP per capita and NIHDI increase over time, so the levels relationship may partly reflect their common time trend.

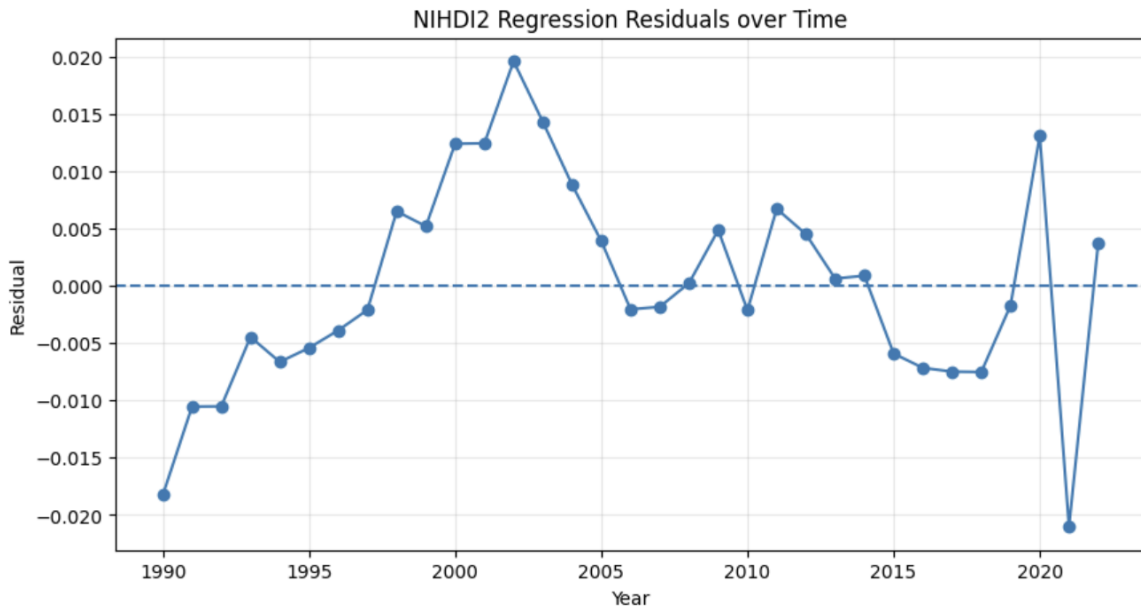
## 4.2 Residual analysis and serial correlation

For each observation, the OLS residual is calculated as:

$$\hat{u}_t = NIHDI2_t - \widehat{NIHDI2}_t.$$

A residual plot over time revealed persistent runs of positive and negative residuals rather than purely random fluctuations around zero. The Durbin–Watson statistic was approximately:

$$DW = 0.999.$$



Since a value near 2 is generally associated with an absence of first-order residual autocorrelation, the value which was very close to 1, indicated substantial positive serial dependence.

The Breusch–Godfrey test was therefore applied formally. Its null hypothesis is:

$$H_0: \rho_1 = 0,$$

where  $\rho_1$  is the first-order residual autocorrelation coefficient. The test produced:

$$p = 0.012.$$

Thus, the null of no first-order serial correlation was rejected at the 5% level. The baseline model therefore left systematic time dependence unexplained.

This diagnostic was important for two reasons. First, it indicated that the very high  $R^2$  could not be interpreted without considering the time-series structure. Second, serially correlated errors (shown visually by the residual plot) can make normal OLS standard errors and hypothesis tests unreliable.

### 4.3 Heteroskedasticity diagnostic

The Breusch–Pagan test was used to examine whether the variance of the regression errors depended systematically on the explanatory variable. Its null hypothesis is:

$$H_0: \text{Var}(u_t) = \sigma^2.$$

The test produced:

$$p \approx 0.607.$$

Therefore, there was no evidence of heteroskedasticity at conventional significance levels in the baseline specification. Heteroskedasticity was consequently not treated as the principal source of concern in the model.

#### 4.4 Stationarity analysis

Because the data are annual time series, Augmented Dickey–Fuller (ADF) tests were conducted. The null hypothesis of the ADF test is that the series contains a unit root, indicating non-stationarity.

An initial ADF exercise using automatic lag selection by AIC produced the following level results:

| Variable                  | <i>p</i> -value | Used lags |
|---------------------------|-----------------|-----------|
| <i>GDPpc</i>              | 0.9888          | 0         |
| <i>NIHDI</i> <sub>1</sub> | 0.6055          | 2         |
| <i>NIHDI</i> <sub>2</sub> | 0.6996          | 1         |
| <i>NIHDI</i> <sub>3</sub> | 0.9950          | 10        |

These results did not reject the unit-root null for any of the four level variables under that initial specification.

The corresponding first differences produced:

$$\begin{aligned}
 p_{\Delta \ln GDPpc} &= 1.19 \times 10^{-6}, \\
 p_{\Delta NIHDI_1} &= 0.7158, \\
 p_{\Delta NIHDI_2} &= 1.42 \times 10^{-12},
 \end{aligned}$$

and

$$p_{\Delta NIHDI_3} = 0.4790.$$

Thus, the stationarity evidence was strongest for log GDP per capita and *NIHDI*<sub>2</sub>, while the alternative NIHDI measures could not be cleanly classified as stationary in first differences using this initial test specification. In particular, the very large lag choice for *NIHDI*<sub>3</sub> reduced the effective sample substantially, illustrating the limitations of applying heavily parameterised unit-root tests to a dataset containing only 33 observations.

Because the variables exhibited pronounced deterministic trends, the ADF tests for the principal specification were subsequently repeated with both a constant and a deterministic time trend. Under this specification:

$$ADF_{\ln GDPpc} = -3.294, p = 0.0673,$$

while:

$$ADF_{NIHDI_2} = -4.242, p = 0.00387.$$

The GDP result is therefore borderline at the 5% level, whereas the unit-root null for  $NIHDI_2$  is rejected when a deterministic trend is included. For the first differences, the p-values remained extremely small:

$$\begin{aligned} p_{\Delta \ln GDP_{pc}} &= 1.19 \times 10^{-6}, \\ p_{\Delta NIHDI_2} &= 1.42 \times 10^{-12}. \end{aligned}$$

These results indicate strong trend and persistence in the levels and provide a statistical basis for considering change-based specifications. They do not, however, justify imposing a definitive cointegration or non-cointegration conclusion, so no strong conclusion is made.

#### 4.5 Regression with an explicit time trend

To determine whether GDP retains a distinct association after accounting for the general progression of time, the following model was estimated:

$$NIHDI2_t = \beta_0 + \beta_1 \ln(GDP_{pc}_t) + \beta_2 t + u_t.$$

The estimated GDP coefficient became:

$$\hat{\beta}_1 = -0.0029, p = 0.927,$$

while the coefficient on the time trend was approximately:

$$\hat{\beta}_2 = 0.0082, p < 0.001.$$

The model's overall explanatory power remained high, with:

$$R^2 = 0.994.$$

However, the variance inflation factor for both log GDP per capita and the trend variable was:

$$VIF \approx 144.6.$$

This demonstrates extreme collinearity between economic income and the passage of time in the sample. The insignificant GDP coefficient should therefore **not** be interpreted as evidence of a negative effect of income. Rather, the result indicates that the sample does not contain sufficient independent variation to distinguish the GDP association from the general development trend.

The time-controlled specification also improved the serial-correlation diagnostics. The Durbin–Watson statistic increased to:

$$DW = 1.531,$$

while the Breusch–Godfrey test gave:

$$p = 0.173.$$

Thus, there was no evidence of first-order serial correlation at the 5% level in this specification. The improvement suggests that part of the serial dependence observed in the initial model was associated with omitted time structure.

Residual normality was less satisfactory in the trend model. The Jarque–Bera test produced:

$$p = 1.11 \times 10^{-27},$$

with skewness of  $-2.312$  and kurtosis of  $11.300$ . This indicates substantial departure from normality. Given the small sample, this further supports cautious interpretation of conventional small-sample inference.

#### 4.6 From levels to annual changes

Because the levels relationship was strongly intertwined with time, the analysis shifted to changes in the variables:

$$\Delta NIHDI_t = NIHDI_t - NIHDI_{t-1}$$

and

$$\Delta \ln(GDPpc_t) = \ln(GDPpc_t) - \ln(GDPpc_{t-1}).$$

The latter is approximately the annual growth rate of GDP per capita. We have applied the log transformations for reasons specified already.

The contemporaneous model was:

$$\Delta NIHDI_{2t} = \alpha + \beta \Delta \ln(GDPpc_t) + \varepsilon_t.$$

The estimated coefficient was:

$$\hat{\beta} = 0.0091,$$

with:

$$p = 0.858, R^2 = 0.001.$$

Hence, same-year GDP-per-capita growth had essentially no explanatory power for same-year improvement in  $NIHDI_2$ . The result does not imply that income growth cannot affect human development; rather, it indicates that a contemporaneous one-year specification does not detect a statistically significant relationship in this sample.

The residuals in this change model were also markedly non-normal. The Jarque–Bera statistic produced:

$$p = 5.49 \times 10^{-17},$$

with kurtosis of approximately  $10.49$ . The Durbin–Watson statistic, however, was:

$$DW = 2.418,$$

indicating no obvious positive first-order serial dependence.

#### 4.7 One-year lagged specification

The absence of a contemporaneous relationship motivated an examination of whether economic growth could take time to translate into health and education related improvements. The final specification was:

$$\Delta NIHDI_t = \alpha + \beta \Delta \ln(GDPpc_{t-1}) + \varepsilon_t$$

where  $\Delta \ln(GDPpc_{t-1})$  represents GDP-per-capita growth in the preceding year.

For  $NIHDI_1$ , the estimated coefficient was:

$$\hat{\beta} = 0.2189,$$

with:

$$p = 0.000163, R^2 = 0.392.$$

For  $NIHDI_2$ :

$$\begin{aligned} \hat{\beta} &= 0.1733, \\ p &= 0.000170, R^2 = 0.391. \end{aligned}$$

For  $NIHDI_3$ :

$$\begin{aligned} \hat{\beta} &= 0.1577, \\ p &= 0.000097, R^2 = 0.413. \end{aligned}$$

All three coefficients are positive and statistically significant at conventional levels. The estimated relationship therefore survives alternative constructions of the non-income index.

For the principal  $NIHDI_2$  specification, the coefficient can be interpreted approximately as follows: because a 0.01 change in  $\Delta \ln(GDPpc)$  corresponds approximately to a one-percentage-point increase in GDP-per-capita growth, a one-percentage-point increase in previous-year GDP-per-capita growth is associated with approximately

$$0.1733(0.01) = 0.001733$$

higher current-year  $NIHDI_2$ , within this simple specification. This should be interpreted strictly as an association and not as a causal marginal effect.

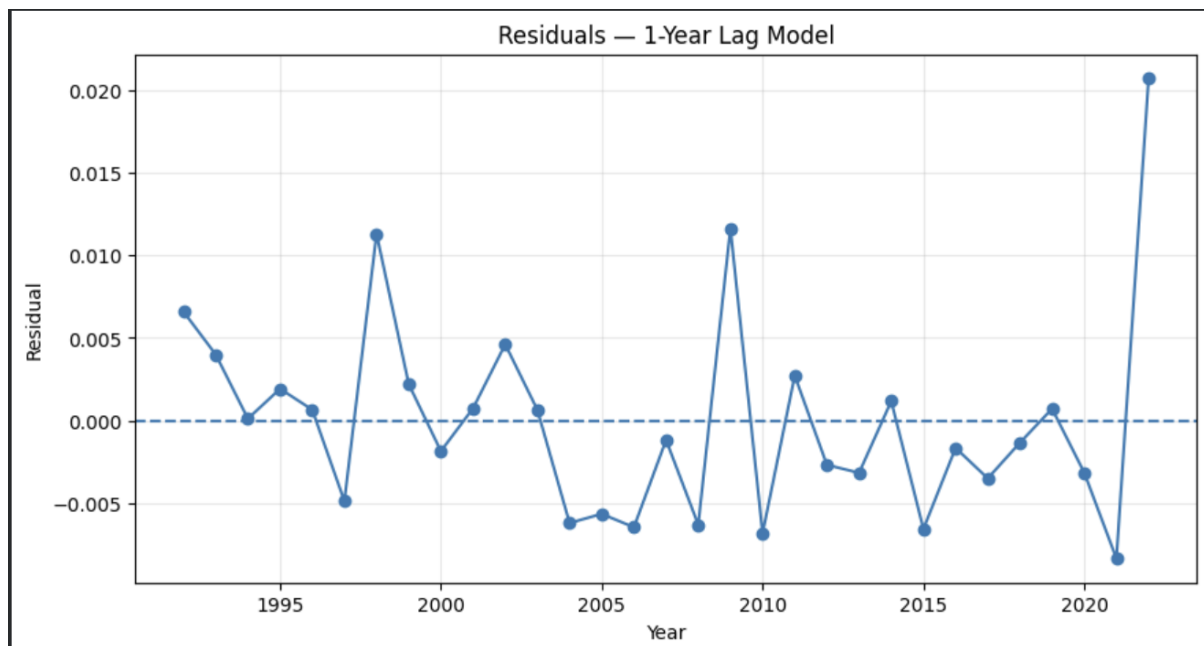
The Durbin–Watson statistics for the three lagged models were:

$$1.976, 1.992, 1.802$$

for  $NIHDI_1$ ,  $NIHDI_2$ , and  $NIHDI_3$ , respectively. The principal  $NIHDI_2$  model had a Breusch–Godfrey p-value of:

$$p = 0.163,$$

providing no evidence of first-order serial correlation at the 5% level. The residual plot also showed considerably less pronounced temporal clustering than the baseline model.



*(generated by the author)*

The final  $NIHDI_2$  model nevertheless displayed non-normal residuals, with a Jarque–Bera p-value of approximately:

0.000223,

skewness of 1.383, and kurtosis of 5.316. Thus, despite the improved serial-correlation behaviour, the small-sample conventional p-values should be interpreted with caution.

## 4.8 Overall statistical interpretation

The sequence of models produces a coherent empirical progression. The initial levels regression reveals an extremely strong GDP–NIHDI association, but its interpretation is complicated by common time trends and serially correlated residuals. Explicitly introducing a time trend shows that GDP and time are too highly collinear to identify a separate GDP coefficient reliably. Moving to annual changes removes much of the common long-run trend but reveals that same-year GDP growth is not associated with same-year NIHDI improvement. Introducing a one-year lag produces a positive and statistically significant association across all three NIHDI constructions.

The central statistical result is therefore not that GDP explains approximately 99% of human development. Rather, the more informative finding is that previous-year GDP-per-capita growth is positively associated with subsequent improvement in non-income human development, and this qualitative relationship is robust to alternative index constructions. Given the observational nature of the data, the small number of annual observations, and the

absence of an explicit causal identification strategy, the estimated relationships should be interpreted as associations rather than causal effects.

### **Policy implications of the study**

- GDP growth should be complemented by human development indicators: economic expansion alone cannot ascertain if gains are translating into improvements in health and education.
- Focus on the conversion of growth into capabilities: strengthen the channels through which higher incomes and fiscal capacity can yield better health and educational outcomes.
- Track non-income outcomes alongside GDP: regular monitoring of health, education and broader capability indicators can reveal whether economic gains are translating into substantive improvements
- Allow for implementation lags: improvements in human development may not appear in the same period as economic growth. policy evaluation should therefore consider delayed outcomes.
- Avoid interpreting growth as an automatic pathway to well being: the positive lagged association in this study does not establish that growth by itself causes human-development gains; the mechanisms behind the association require further investigation.
- Use alternative measurement frameworks as robustness checks: the persistence of the lagged relationship across all three NIHDI constructions suggests that conclusions should not rest on a single composite-index specification. (Which was also the initial assumption of the study)

### **Limitations**

The analysis is based on only 33 annual observations for a single country. That limits the complexity of the models and the extent to which the findings can be generalised. More importantly, the data are observational, so the positive lagged association can be interpreted only as an association and not as a causal effect of GDP growth on human development. Health and education outcomes are also shaped by factors such as public policy, access, institutions and inequality, none of which are explicitly modelled here. The results should therefore be read as evidence of a statistical relationship and its timing, rather than as a complete explanation of how economic growth is converted into human-development gains.

### **Conclusion**

The analysis suggests that economic growth and non-income human development have moved closely together over India's development trajectory, but the simple levels relationship cannot by itself tell us whether higher GDP is independently associated with better human-development outcomes. The very high raw  $R^2$  is strongly intertwined with the common upward time trend in both variables, while the time-controlled specification shows that GDP and the broader passage of time are difficult to disentangle in this sample. Moving from levels to annual changes therefore provides a more informative perspective: same-year GDP growth shows no detectable association with improvement in non-income human development, whereas previous-year GDP growth is positively associated with subsequent improvement in NIHDI. Importantly, this result is remarkably consistent across all three alternative NIHDI constructions, suggesting that it is not simply a consequence of one particular way of measuring non-income human development.

The broader implication is that economic growth and human development should not be treated as synonymous. Growth may create the resources and fiscal capacity through which improvements in health and education become possible, but the relationship is neither instantaneous nor necessarily automatic. The presence of a lag points towards a process through which economic gains are translated into broader human capabilities over time. Thus, GDP remains relevant to development, but GDP alone is insufficient for understanding whether economic progress is actually improving people's substantive well-being.

## **References**

- Akaike, H. (1974). A new look at the statistical model identification. *IEEE Transactions on Automatic Control*, 19(6), 716–723.
- Anand, S., & Sen, A. (2000). The income component of the Human Development Index. *Journal of Human Development*, 1(1), 83–106.
- Belsley, D. A., Kuh, E., & Welsch, R. E. (1980). *Regression diagnostics: Identifying influential data and sources of collinearity*. Wiley.
- Breusch, T. S. (1978). Testing for autocorrelation in dynamic linear models. *Australian Economic Papers*, 17(31), 334–355.
- Breusch, T. S., & Pagan, A. R. (1979). A simple test for heteroscedasticity and random coefficient variation. *Econometrica*, 47(5), 1287–1294.
- Crafts, N. F. R. (1997a). The Human Development Index and changes in standards of living: Some historical comparisons. *European Review of Economic History*, 1(3), 299–322.
- Crafts, N. F. R. (1997b). Some dimensions of the "quality of life" during the British Industrial Revolution. *Economic History Review*, 50(4), 617–639.
- Dickey, D. A., & Fuller, W. A. (1979). Distribution of the estimators for autoregressive time series with a unit root. *Journal of the American Statistical Association*, 74(366), 427–431.

- Drèze, J., & Sen, A. (2013). *An uncertain glory: India and its contradictions*. Princeton University Press.
- Durbin, J., & Watson, G. S. (1950). Testing for serial correlation in least squares regression: I. *Biometrika*, 37(3–4), 409–428.
- Durbin, J., & Watson, G. S. (1951). Testing for serial correlation in least squares regression: II. *Biometrika*, 38(1–2), 159–177.
- Engle, R. F., & Granger, C. W. J. (1987). Co-integration and error correction: Representation, estimation, and testing. *Econometrica*, 55(2), 251–276.
- Godfrey, L. G. (1978). Testing against general autoregressive and moving average error models when the regressors include lagged dependent variables. *Econometrica*, 46(6), 1293–1301.
- Granger, C. W. J., & Newbold, P. (1974). Spurious regressions in econometrics. *Journal of Econometrics*, 2(2), 111–120.
- Jarque, C. M., & Bera, A. K. (1987). A test for normality of observations and regression residuals. *International Statistical Review*, 55(2), 163–172.
- Kuznets, S. (1934). *National income, 1929–1932* (73rd Congress, 2nd Session, Senate Document No. 124). U.S. Government Printing Office.
- Prados de la Escosura, L. (2022). Augmented human development in the age of globalization. *Economic History Review*.
- Ranis, G., Stewart, F., & Ramirez, A. (2000). Economic growth and human development. *World Development*, 28(2), 197–219.
- Rijpma, A., Philips, R. C. M., & van Bavel, B. J. P. (2025). Multidimensional composite indicators of well-being: Applications in economic history. *Journal of Economic Surveys*, 39(2), 672–705.
- Said, S. E., & Dickey, D. A. (1984). Testing for unit roots in autoregressive-moving average models of unknown order. *Biometrika*, 71(3), 599–607.
- Seabold, S., & Perktold, J. (2010). Statsmodels: Econometric and statistical modeling with Python. *Proceedings of the 9th Python in Science Conference*, 92–96.
- Sen, A. (1985). *Commodities and capabilities*. North-Holland.
- Sen, A. (1999). *Development as freedom*. Oxford University Press.
- Stiglitz, J. E., Sen, A., & Fitoussi, J.-P. (2009). *Report by the Commission on the Measurement of Economic Performance and Social Progress*.  
<https://www.insee.fr/en/information/2662494>

United Nations Development Programme. (1990). *Human development report 1990: Concept and measurement of human development*. Oxford University Press.

United Nations Development Programme. (2010). *Human development report 2010: The real wealth of nations: Pathways to human development*. Palgrave Macmillan.

World Bank. (n.d.). *World development indicators* [Data set].  
<https://databank.worldbank.org/source/world-development-indicators>.

## **Appendix:**

Link to the Github repository: <https://github.com/gangulyabhishikta36-lang/NIHDI-PROJECT>

(the following repository contains all the code files associated with the project)